FedKG-Maint: Privacy-Preserving Predictive Maintenance

Predictive maintenance uses machine sensor data to predict failures before they happen, reducing downtime and cost. However, a single factory usually has too few failure examples to train a reliable model, and factories cannot share their raw data because of privacy and business confidentiality. In this project, students will build a framework that allows multiple factories to learn a shared predictive maintenance model using federated learning, without sharing their raw data.

The project will also address explainability: maintenance engineers need to understand why a failure is predicted and what action to take. Therefore, the framework will combine a maintenance knowledge graph with a Large Language Model (LLM) to produce clear, grounded explanations and recommendations, and will examine whether the approach generalizes to factories with different machines and operating conditions.

Key Steps :

1. Data Collection and Preparation

- Use the NASA C-MAPSS and Microsoft Azure Predictive Maintenance datasets, which contain sensor telemetry, operating conditions, error logs, maintenance records, and failure events of industrial machines.
- Simulate multiple factories by splitting the data into separate clients, for example using the four C-MAPSS sub-datasets (FD001–FD004), which differ in operating conditions and fault modes, or grouping the Azure machines by model.
- Make sure the factories are non-identically distributed, and keep a held-out factory that is never used for training, to test generalization to a new site.

2. Data Preprocessing and Feature Engineering

- Clean and normalize the sensor signals, handle missing values, and remove constant or uninformative sensors.
- Create time windows and prediction targets, such as Remaining Useful Life (RUL) or failure within the next N hours.
- Normalization statistics must be computed locally inside each factory, since raw data cannot be shared.

3. Maintenance Knowledge Graph Construction

- Build a knowledge graph that links machines, components, sensors, error codes, failure modes, and maintenance actions.
- Each factory builds its own local knowledge graph from its data, while a shared schema (ontology) is agreed on by all factories.
- Optionally, enrich the graph with general domain knowledge, such as known relations between failure modes and their symptoms.

4. Federated Learning for Predictive Maintenance

- Train a predictive model (e.g., LSTM, Transformer, or Graph Neural Network over the knowledge graph) using federated learning, so that only model updates, not raw data, leave each factory.
- Compare at least two federated strategies (e.g., FedAvg and FedProx or a personalized FL method) against two reference setups: local-only training in each factory and centralized training on pooled data.
- Evaluate on the held-out factory to measure cross-factory generalization.

5. Privacy Preservation

- Apply at least one privacy-preserving mechanism, such as differential privacy on model updates or secure aggregation.
- Analyze the trade-off between the privacy level and prediction performance.
- Discuss what information could still leak through the model updates or the shared knowledge graph.

6. LLM-Based Explainable Maintenance

- Use a Large Language Model together with the knowledge graph (e.g., retrieval of relevant graph facts) to generate human-readable explanations and maintenance recommendations for each prediction.
- Explanations must be grounded in the model output and the knowledge graph facts, not invented by the LLM.
- Compare LLM explanations with and without knowledge graph grounding, and check them for factual correctness and hallucinations.

7. Evaluation and Analysis

- Evaluate prediction performance using suitable metrics (e.g., RMSE and the NASA scoring function for RUL on C-MAPSS, F1-score and PR-AUC for failure prediction on the Azure dataset).
- Measure communication cost and the effect of the number of factories and the degree of data heterogeneity.
- Evaluate explanation quality through faithfulness and correctness checks, and a small human evaluation if possible.
- Run ablation studies (e.g., without the knowledge graph, without privacy, without the LLM) and report results over multiple runs with statistical significance testing.
- Discuss the main findings and limitations of the developed framework.

Datasets

- **NASA C-MAPSS (Turbofan Engine Degradation):** https://data.nasa.gov/dataset/c-mapss-aircraft-engine-simulator-data
- **Microsoft Azure Predictive Maintenance:** https://www.kaggle.com/datasets/arnabbiswas1/microsoft-azure-predictive-maintenance